

习题十

2020 年 12 月 08 日

1. 第 10 章练习题第 2 题。
2. 第 10 章练习题第 6 题。
3. 用 Spice 仿真 smic18 单元库中的带异步复位端的 D 触发器 (DFFRX1) 电路的建立时间和传播延时。

单元电路网表如下:

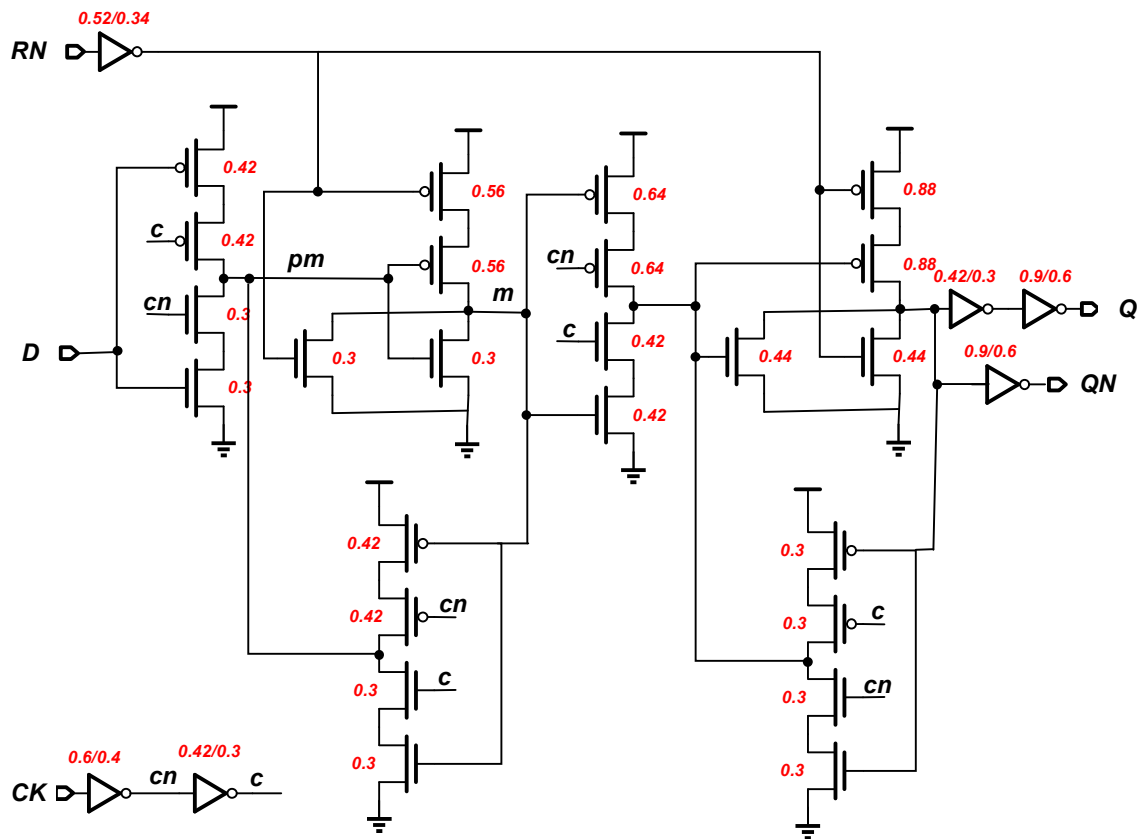
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.subckt DFFRX1 Q QN CK D RN
M0 VDD net52 hnet17 VDD P18 l=0.18u w=0.88u
M1 hnet17 net67 s VDD P18 l=0.18u w=0.88u
M2 s net52 VSS VSS N18 l=0.18u w=0.44u
M3 s net67 VSS VSS N18 l=0.18u w=0.44u
M4 VDD net52 hnet23 VDD P18 l=0.18u w=0.56u
M5 hnet23 pm m VDD P18 l=0.18u w=0.56u
M6 m net52 VSS VSS N18 l=0.18u w=0.3u
M7 m pm VSS VSS N18 l=0.18u w=0.3u
M8 net46 s VSS VSS N l=0.18u w=0.3u
M9 VDD s net46 VDD P18 l=0.18u w=0.42u
M10 QN s VSS VSS N18 l=0.18u w=0.6u
M11 VDD s QN VDD P18 l=0.18u w=0.9u
M12 Q net46 VSS VSS N18 l=0.18u w=0.6u
M13 VDD net46 Q VDD P18 l=0.18u w=0.9u
M14 net52 RN VSS VSS N18 l=0.18u w=0.34u
M15 VDD RN net52 VDD P18 l=0.18u w=0.52u
M16 cn CK VSS VSS N18 l=0.18u w=0.4u
M17 VDD CK cn VDD P18 l=0.18u w=0.6u
M18 c cn VSS VSS N18 l=0.18u w=0.3u
M19 VDD cn c VDD P18 l=0.18u w=0.42u
M20 hnet43 cn net67 VDD P18 l=0.18u w=0.64u
M21 VDD m hnet43 VDD P18 l=0.18u w=0.64u
M22 hnet41 m VSS VSS N18 l=0.18u w=0.42u
M23 net67 c hnet41 VSS N18 l=0.18u w=0.42u
M24 hnet49 c pm VDD P18 l=0.18u w=0.42u
M25 VDD D hnet49 VDD P18 l=0.18u w=0.42u
M26 hnet47 D VSS VSS N18 l=0.18u w=0.3u
M27 pm cn hnet47 VSS N18 l=0.18u w=0.3u
M28 hnet55 c net67 VDD P18 l=0.18u w=0.3u
M29 VDD s hnet55 VDD P18 l=0.18u w=0.3u
M30 hnet53 s VSS VSS N18 l=0.18u w=0.3u
M31 net67 cn hnet53 VSS N18 l=0.18u w=0.3u
M32 hnet61 cn pm VDD P18 l=0.18u w=0.42u
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M33 VDD m hnet61 VDD P18 l=0.18u w=0.42u
M34 hnet59 m VSS VSS N18 l=0.18u w=0.3u
M35 pm c hnet59 VSS N18 l=0.18u w=0.3u
.ends DFFRX1

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对应的电路图如下：



单元说明文档如下：

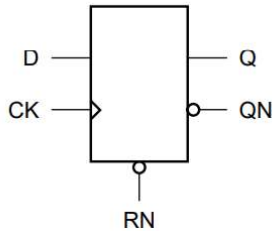
Cell Description

The DFFR cell is a positive-edge triggered, static D-type flip-flop with asynchronous active-low reset (RN).

Functions

RN	D	CK	Q[n+1]	QN[n+1]
0	x	x	0	1
1	0		0	1
1	1		1	0
1	x		Q[n]	QN[n]

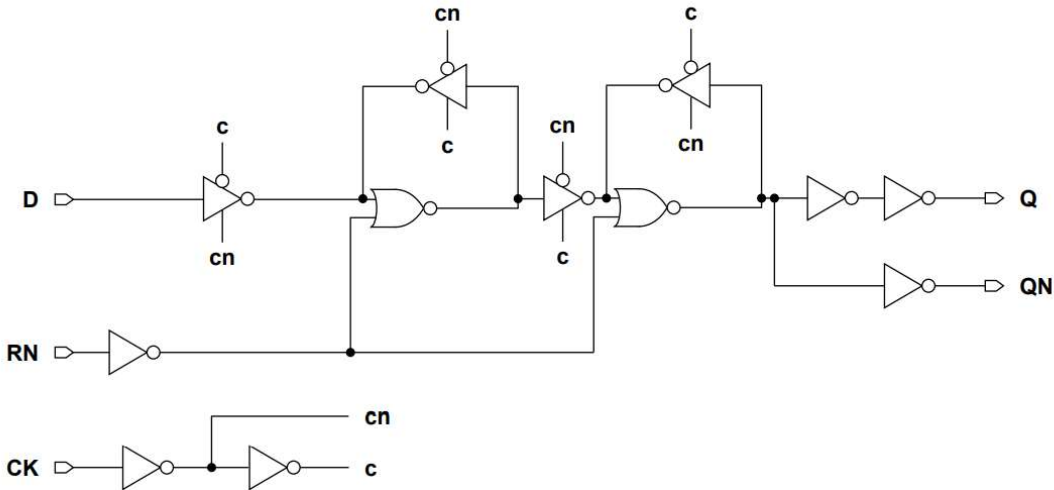
Logic Symbol



Cell Size

Drive Strength	Height (μm)	Width (μm)
DFFRXL	5.04	15.18
DFFRX1	5.04	15.18
DFFRX2	5.04	17.16
DFFRX4	5.04	19.80

Functional Schematic



AC Power

Pin	Power (μ W/MHz)			
	XL	X1	X2	X4
D	0.0356	0.0286	0.0326	0.0428
CK	0.0370	0.0351	0.0387	0.0452
RN	0.0160	0.0180	0.0221	0.0341
Q	0.0500	0.0563	0.0897	0.1548

Pin Capacitance

Pin	Capacitance (pF)			
	XL	X1	X2	X4
D	0.0035	0.0024	0.0022	0.0031
CK	0.0025	0.0030	0.0031	0.0042
RN	0.0023	0.0027	0.0034	0.0061

Delays at 25°C, 1.8V, Typical Process

Description	Intrinsic Delay (ns)				K_{load} (ns/pF)			
	XL	X1	X2	X4	XL	X1	X2	X4
CK $\rightarrow$ Q $\uparrow$	0.4355	0.3976	0.3961	0.3536	5.8906	4.2015	2.0803	1.0643
CK $\rightarrow$ Q $\downarrow$	0.3013	0.3186	0.3046	0.2774	3.4685	2.5313	1.2943	0.6517
RN $\rightarrow$ Q $\downarrow$	0.2477	0.2497	0.2405	0.2183	3.4691	2.5314	1.2943	0.6517
CK $\rightarrow$ QN $\uparrow$	0.2514	0.2543	0.2307	0.2098	5.8878	4.2019	2.0814	1.0649
CK $\rightarrow$ QN $\downarrow$	0.4065	0.3601	0.3431	0.2986	3.7020	2.5945	1.3188	0.6627
RN $\rightarrow$ QN $\uparrow$	0.1980	0.1862	0.1671	0.1511	5.8994	4.2059	2.0835	1.0662

Timing Constraints at 25°C, 1.8V, Typical Process

Pin	Requirement	Interval (ns)			
		XL	X1	X2	X4
D	setup $\uparrow$ $\rightarrow$ CK	0.0547	0.0781	0.0977	0.0781
	setup $\downarrow$ $\rightarrow$ CK	0.0859	0.1406	0.1680	0.1406
	hold $\uparrow$ $\rightarrow$ CK	-0.0391	-0.0586	-0.0664	-0.0547
	hold $\downarrow$ $\rightarrow$ CK	0.0273	-0.0391	-0.0352	-0.0312
CK	minpwh	0.1321	0.1418	0.1321	0.1127
	minpwl	0.1953	0.1759	0.2001	0.1564
RN	minpwl	0.1807	0.1807	0.2050	0.2973
	recovery	0.0430	0.0508	0.0703	0.0547
	removal	-0.0234	-0.0352	-0.0430	-0.0195